

Unique challenges of the Indian EV market

The Indian EV market, especially in the micro-mobility segment, faces distinct challenges due to its unique environment and consumer preferences. Safety concerns are heightened as these smaller vehicles, such as scooters, two-wheelers, and three-wheelers, are often charged inside homes, increasing the risk to owners in the event of a battery issue. Key challenges include:

Prevalence of micro-mobility. Scooters and two-wheelers are more common, leading to different safety and infrastructure needs compared to larger EVs

Battery quality concerns. Lower-cost vehicles have raised questions about battery quality, prompting government regulations to ensure safety

High ambient temperatures. With temperatures reaching 40-50°C, the lack of active cooling strategies in these vehicles poses a significant challenge for thermal management

Cost sensitivity. Providing adequate protection while meeting the low price points expected in the Indian market is a major hurdle

Despite these challenges, there is a push towards innovative solutions like encapsulation and phase change materials to improve thermal management without active cooling systems. Companies such as Ola, Pure EV, and Okinawa have faced recalls due to fire risks, highlighting the urgent need for enhanced safety measures. As the market evolves, ensuring the safety and reliability of EVs in India's unique conditions will be crucial for widespread adoption and consumer trust.

critical. The expanding EV market has prompted companies to adapt existing products or develop new ones specifically designed for EV safety. This burgeoning sector has attracted a growing number of players, all vying for a share of the market and aiming to meet the rising demand for effective fire protection solutions in EVs.

Battery cell formats and design flexibilities

In the EV industry, the choice of battery cell formats is a critical consideration, with prismatic, cylindrical, and pouch cells being the primary options. Prismatic cells are favoured for their ease of stacking and suitability for incorporating insulating materials. This makes them a popular choice in markets like China. Major cell suppliers such as CATL have adopted this format, while others like LG Chem prefer the pouch format for its superior energy density.

Pouch cells, despite their higher energy density, present challenges in pack construction due to their expansion and contraction during operation and across their lifespan. Manufacturers like GM have devel-

oped battery packs that can accommodate different cell formats, including prismatic and pouch cells, with cylindrical cells announced for the future, providing flexibility in design and sourcing. Cylindrical cells, on the other hand, offer more flexibility in pack design but require more individual connections, making the pack structure more complex.

Tesla's shift from smaller cylindrical cells to larger ones has simplified manufacturing and enabled a cell-to-pack approach, influencing the industry's adoption of cylindrical cells. However, each cell format has its advantages and disadvantages, affecting the design and performance of EV battery packs. Prismatic cells are easier to construct into a pack, cylindrical cells provide flexibility in design, and pouch cells offer the highest energy density but require careful management of their expansion and contraction.

EV manufacturers are exploring various strategies to accommodate different cell types, with trends towards cell-to-pack designs that eliminate modules, reducing materials and complexity. This diversity in

cell formats allows for more options in battery pack design and supply but also presents challenges in designing battery packs for different models. Ultimately, the choice of cell format depends on the specific requirements of the EV, including energy density, safety, and manufacturing efficiency.

Fire safety in EV batteries

The primary objective of fire protection materials in EV batteries is to prevent or delay the spread of thermal runaway events within the battery pack, rather than merely extinguishing fires. Once a cell undergoes thermal runaway, the focus shifts to containing the event and preventing it from affecting adjacent cells. The choice of materials for this purpose varies depending on the cell type and design considerations. Historically, ceramic blankets and mica have been popular choices for their fire protection properties, despite some drawbacks such as dustiness and brittleness, respectively.

Recent trends have seen a shift towards alternative materials like aerogels, which offer the dual benefits of fire protection and thermal insulation while being lighter than traditional options. In battery packs with cylindrical cells, encapsulating foams are commonly used to provide insulation and structural support, as well as to reduce the oxygen available for combustion within the pack.

The role of encapsulating foams extends beyond insulation, as they also contribute to electrical safety by preventing arcing and the ignition of vented gases. There is a tactical shift towards designs that eliminate the module housing, necessitating fire protection materials that can conform to the shape of the cells and provide effective thermal insulation.

Aerogels are particularly appealing in this context due to their excellent thermal insulation properties and lightweight nature, which can